Venkata Lakshmi Parimala Pasupuleti

AI Engineer · LangGraph · RAG Pipelines · LLM Evaluation · Multi-Agent Systems · Python · FastAPI · AWS

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala/AgentProbe | McLean area, Virginia

PROFESSIONAL SUMMARY

AI Engineer with 8+ years of production Python engineering and hands-on agentic AI system design. Built **AgentProbe** - an open-source multi-agent AI testing toolkit: **3 LangGraph agents** (Orchestrator, RAG Tracer, Auto Fix), **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance), multi-provider LLM routing (Anthropic · OpenAI · Groq · Ollama), ChromaDB RAG pipeline, **15-endpoint FastAPI backend** - 100% local at zero cost. Built **Leasel** - agentic RAG compliance pipeline achieving **93.6% precision** and 87.9% F1 across 10 compliance rules (LangGraph, deterministic rules engine, LLM-as-judge scoring, pass@k eval harness). At Entrata, shipped **production AI pipeline on 1.6M real records** (PyTorch, OpenAI API, output validation, quality gates) and 20+ full-stack modules at scale (**1M+ records, 100K+ MAU**, 99.5% uptime). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

AI PROJECTS

AgentProbe: Agentic AI Testing Toolkit

MULTI-AGENT LLMOPS RAG

Python · LangGraph · LangChain · FastAPI · ChromaDB · LLM-as-Judge · OpenAI · Anthropic · Groq · Ollama · React · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Built a **3-agent LangGraph runtime** (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama) — 15-endpoint FastAPI backend running 100% locally at zero cost.
- Blocked failing agent runs before production by building a **5-metric LLM-as-judge eval engine** (factual accuracy, hallucination, retrieval, completeness, compliance) with **CI/CD integration** via GitHub Actions — every commit evaluated automatically; regression drift caught before merge.
- Delivered trace-level observability across concurrent agent test suites — per-run metrics, retrieval quality, latency, and regression drift — via a React dashboard backed by **ChromaDB RAG pipeline** with semantic search over agent trace history.

Leasel: Agentic Compliance Pipeline

AGENTIC RAG LLM-AS-JUDGE

Python · LangGraph · SQLite · Deterministic Rules Engine · Pass@k Eval — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Achieved **93.6% precision and 87.9% F1** across 10 compliance rules on 1,240 residential lease PDFs by building an agentic LangGraph RAG pipeline — 5 US states + federal standards, 1,050 synthetic tenants, ~40% injected violations; deterministic rules engine keeps all compliance determinations fully auditable.
- Built a **pass@k evaluation harness** (pass@1 81.7%, pass@3 88.3%) with multi-trial sampling, regression gate, and LLM-as-judge scoring across accuracy and reasoning quality — demonstrates production-grade eval design for agentic workflows.

CrashClarity: Large-Scale ML Analytics (104K Records)

ML

Python · R · Scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages — Live Dashboard

- Achieved **85%+ accuracy** on 104K+ UK accident records by building a Random Forest + Logistic Regression ensemble with feature engineering across weather, road, and time variables; deployed interactive dashboard on GitHub Pages.

WORK EXPERIENCE

Drone Mapping Startup

May 2025 – Aug 2025

AI/ML Engineer Intern — Agentic Mission Planning · RAG · Geospatial LLM Pipelines · LangGraph

- Built a **LangGraph agentic mission planning pipeline** that parsed natural-language flight intent into structured waypoints and MAVLink-compatible flight parameters — validated plans against battery budget, FAA Part 107 airspace constraints, LAANC authorizations, and weather data before generating any flight path; LLMs applied at the planning/reasoning layer, not in the real-time flight loop.
- Built **RAG pipeline (ChromaDB)** over FAA Part 107 regulations, airspace classification documents, NOTAM data, and manufacturer manuals — natural-language compliance and troubleshooting queries; paired text retrieval with structured airspace API lookups to prevent hallucination on safety-critical altitude and legality decisions.
- Applied **multimodal pipeline over orthomosaic mapping outputs**: YOLO-based detection for aerial feature identification, VLM layer for contextualizing findings and generating structured inspection reports (detected features + geotags + severity scores); implemented conversational geospatial querying over survey datasets via LLM-to-SQL (GDAL/GeoPandas) for non-technical stakeholder access.

George Mason University

Aug 2024 – Aug 2025

Research Associate — ML Pipelines · Anomaly Detection · Statistical Modeling · AI Automation

- Applied **Isolation Forest anomaly detection** across 3 research pipelines (Python) — surfaced statistical outliers that manual review missed; built end-to-end ETL + modeling pipelines on multi-source datasets (WHO + UK Road Safety, 190+ countries) achieving **$R^2 = 0.977$**.
- Automated statistical reporting pipelines (Python, Quarto) championing generative AI adoption in research workflows; designed faculty leadership dashboards (Tableau, ggplot2) improving comprehension **40%** across two cohorts.

Entrata Jan 2021 – Aug 2024

Senior Software Engineer — Production AI Pipelines · AWS Data Infrastructure · Python at Scale

- Shipped Entrata's **first production AI pipeline on 1.6M real financial records** — PyTorch, OpenAI API, output validation, quality gates, and compliance controls in place at launch; feature later productized into a customer-facing product.
- Eliminated **80% of manual operational workload** by building Python + PL/SQL automation pipelines across income eligibility verification, AMI threshold calculations, and recertification workflows covering 100K+ unit portfolios.
- Migrated all client data to **AWS (S3, RDS, Glue)** with encrypted ETL pipelines achieving **100% PHI compliance** and zero security incidents; optimized **200+ SQL queries for 30% performance improvement** across 1M+ records at **99.5% uptime**.

Entrata Jan 2019 – Dec 2020

Software Engineer — Data Pipelines · Compliance Automation · Python

- Built and maintained compliance automation pipelines (Python, PostgreSQL) — encoded complex business logic into deterministic rule-based systems processing millions of tenant certifications with full data lineage and audit trails.

Entrata Jan 2017 – Dec 2018

Associate Engineer — Full-Stack Development · Python · PostgreSQL · REST APIs

- Built and maintained full-stack features and SQL data layers (PHP, JavaScript, PostgreSQL) for property management workflows; tuned queries across client database environments supporting 100K+ unit portfolios.

EDUCATION

M.S. Data Analytics Engineering — George Mason University Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · ML systems, statistical modeling, data engineering, AI pipelines, NLP, data governance

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur 2013 – 2017

TECHNICAL SKILLS

AI & Agents: LangGraph, LangChain, Multi-Agent Orchestration, Agentic AI, Tool Calling, RAG Pipelines, ChromaDB, Vector Databases, Semantic Search, Embeddings, LLM-as-Judge, LLM Evaluation, Eval Frameworks, Agent Observability, Agent Tracing, Prompt Engineering, OpenAI API, Anthropic API, Groq, Ollama, FastAPI, REST APIs

ML & Data Science: PyTorch, Scikit-learn, XGBoost, Random Forest, Logistic Regression, Isolation Forest, Anomaly Detection, YOLO, Vision-Language Models (VLM), Multimodal AI, Feature Engineering, Statistical Modeling, NLP, BERT, Hugging Face, spaCy, Embeddings, Pandas, NumPy, Jupyter

Geospatial & Drone AI: GDAL, GeoPandas, Rasterio, Orthomosaic Analysis, Geospatial LLM Pipelines, Natural-Language Mission Planning, MAVLink, Aerial Inspection Automation

Languages & Backend: Python, JavaScript, TypeScript, SQL, PL/SQL, R, Bash, PHP; Docker, CI/CD (GitHub Actions), Microservices, RabbitMQ

Cloud & Data Infrastructure: AWS (S3, EC2, RDS, Glue), ETL Pipelines, Data Pipelines, Data Modeling, PostgreSQL, MySQL, Oracle, SQLite, PHI/HIPAA Compliance, Audit Pipelines, Git, Linux